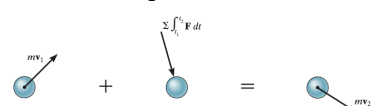
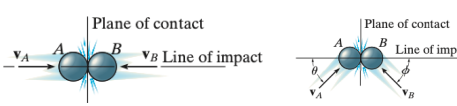
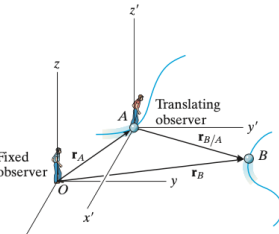
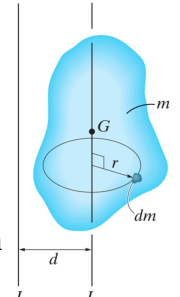


Work of Force $W = F \cdot \Delta s$ $W = F \cdot \Delta s \cdot \cos(\theta)$ $W_{\text{total}} = \sum W$	Work & Energy $K_1 + \sum W_{1 \rightarrow 2} = K_2$ $K = \frac{1}{2}mv^2$ $U_{\text{elastic}} = \frac{1}{2}k\Delta s^2$ $U_{\text{grav}} = mgh$ $U = U_{\text{elastic}} + U_{\text{gravity}}$	Power & Efficiency Power = Fv $P_{\text{avg}} = \frac{\Delta W}{\Delta t}$ Efficiency(ϵ) = $\frac{P_{\text{out}}}{P_{\text{in}}}$ $1W = 1 \frac{J}{s} = 1 \frac{N \cdot m}{s} = 1 \frac{kg \cdot m^2}{s^3}$ $1hp = 550 \frac{ft \cdot lb}{s} = 746w$	Principle of Impulse and Momentum (you can break into x, y or z) $mv_{(xyz)_1} + \sum \int_{t_1}^{t_2} F_{xyz} dt = mv_{(xyz)_2}$ 	
Work on Variable Force $W = \int_{s_i}^{s_f} F \cos(\theta) ds$	Conservation on energy $\sum K_1 + \sum U_1 = \sum K_2 + \sum U_2$	Gears in Mesh $v_A \times r_A = \omega_B \times r_B$ $v_A = \omega_A r_A = \omega_B r_B = v_B$ $\omega_B = \omega_A \left(\frac{r_A}{r_B} \right) = \omega_A \left(\frac{N_{A\text{teeth}}}{N_{B\text{teeth}}} \right)$ If given α_A $\omega_A = \int \alpha_A dt$ 	Impulse $SI: I = N \cdot s$ Variable $I = \int_{t_1}^{t_2} F(t) dt$ Constant $I = F_c(t_2 - t_1)$	Conservation Linear Momentum $\sum m_i v_i = \sum m_f v_f$ Momentum $SI: p = kg \cdot \frac{m}{s}$ Linear $p = mv$
Impact $m_A v_{A_i} + m_B v_{B_i} = m_A v_{A_f} + m_B v_{B_f}$ $v_{A_x} = v_A \cdot \cos(\theta)$ 	Curvilinear Motion (n.t) $a_t = \frac{dv}{dt}$ $a_n = \frac{v^2}{\rho}$ $a_t ds = v dv$ $\vec{a} = a_t \hat{u}_t + a_n \hat{u}_n$ $ \vec{a} = \sqrt{a_t^2 + a_n^2}$ $\rho = \frac{\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{3/2}}{\left \frac{d^2y}{dx^2} \right }$			
Angular Impulse & Momentum $\vec{H}_O = \vec{r} \times m\vec{v} \rightarrow H_{O_z} = (r)(mv)$ $\vec{H}_O = \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ r_x & r_y & r_z \\ mv_x & mv_y & mv_z \end{bmatrix} = H_x \hat{i} + H_y \hat{j} + H_z \hat{k}$ $\sum (H_O)_i + \sum \int_{t_1}^{t_2} M_O dt = \sum (H_O)_f$ $\int_{t_1}^{t_2} \vec{M}_O dt = \int_{t_1}^{t_2} (\vec{r} \times \vec{F}) dt$		Relative Motion (can break into x y z) $\vec{r}_B = \vec{r}_A + \vec{r}_{B/A}$ $\vec{v}_B = \vec{v}_A + \vec{v}_{B/A}$ $\vec{a}_B = \vec{a}_A + \vec{a}_{B/A}$ 		
Kinematics $a = \frac{dv}{dt} \quad v = \frac{ds}{dt}$ $a ds = v dv$ Constant a $v_f = v_o + a_c t$ $s_f = s_o + v_o t + \frac{1}{2} a_c t^2$ $v^2 = v_o^2 + 2a_c(s - s_o)$	Rotation about fixed point $\omega = \frac{d\theta}{dt} \hat{u}_z$ $\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2} \hat{u}_z$ $\alpha d\theta = \omega d\omega$ $\vec{v} = \vec{\omega} \times \vec{r}$ $\vec{a} = \vec{\alpha} \times \vec{r} + \vec{\omega} \times (\vec{\omega} \times \vec{r})$	Kinematic Equations for Rotational Motion Constant α $\omega = \omega_0 + \alpha_c t$ $\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha_c t^2$ $\omega^2 = \omega_0^2 + 2 \alpha_c (\theta - \theta_0)$ $a_t = \alpha r \quad a_n = \omega^2 r$ $v = \omega r$		
Planar Rigid Body Kinetics Moment of Inertia $I = \int r^2 dm$ Radius of Gyration $I = m \cdot k^2$ Parallel Axis Theorem $I = I_G + md^2$ 		Planar equations of Motion $\sum F_x = m(a_G)_x$ $\sum F_y = m(a_G)_y$ $\sum M_G = 0$ $\sum F_n = m(a_G)_n = m\omega^2 r_G$ $\sum F_t = m(a_G)_t = m\alpha r_G$ $\sum M_G = I_G \alpha \text{ or } \sum M_O = I_O \alpha$		
Relative Motion Using Rotating Axes $\vec{v}_B = \vec{v}_A + \vec{\omega} \times \vec{r}_{B/A} + \left(\vec{v}_{B/A} \right)_{xyz}$ $\vec{a}_B = \vec{a}_A + \vec{\alpha} \times \vec{r}_{B/A} + \vec{\omega} \times (\vec{\omega} \times \vec{r}_{B/A}) + 2 \left(\vec{\omega} \times \vec{v}_{B/A} \right)_{xyz} + \left(\vec{a}_{B/A} \right)_{xyz}$		Unit Conversions $1\text{in} = 2.54\text{cm}$ $1\text{mile} = 5280\text{ft}$ $1\text{ m/s} \approx 3.281\text{ ft/s} \quad 1\text{ mph} \approx 1.467\text{ ft/s}$ $1\text{ lb} = 4.448\text{ N}$ $1\text{ slug} = 32.174\text{ lb} \cdot \text{s}^2/\text{ft}$ $g = 9.81\text{ m/s}^2 = 32.174\text{ ft/s}^2$		

Curvilinear Motion (r,θ)

$$v_r = \dot{r} \quad v_\theta = r\dot{\theta}$$

$$\vec{v} = v_r \hat{u}_r + v_\theta \hat{u}_\theta$$

$$|\vec{v}| = \sqrt{v_r^2 + v_\theta^2}$$

$$a_r = \ddot{r} - r\dot{\theta}^2$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$$

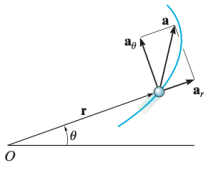
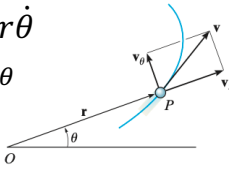
$$\vec{a} = a_r \hat{u}_r + a_\theta \hat{u}_\theta$$

$$|\vec{a}| = \sqrt{a_r^2 + a_\theta^2}$$

Converting between (n,t) and (r,θ)

$$\tan(\psi) = \frac{r}{dr/d\theta}$$

Ψ is the angle between $\hat{u}_t$ & $\hat{u}_r$



Curvilinear Motion (r,θ)

$$\dot{x} = \frac{dx}{dt} = \frac{dx}{dy} \cdot \frac{dy}{dt}$$

$$\dot{y} = \frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$$

$$v_x = \dot{x} \quad a_x = \dot{v}_x$$

$$v_y = \dot{y} \quad a_y = \dot{v}_y$$

$$v_z = \dot{z} \quad a_z = \dot{v}_z$$

$$v = \sqrt{v_x^2 + v_y^2}$$

Projectile Motion

$$x = x_0 + (v_0)_x t$$

$$v_y = (v_0)_y + a_c t$$

$$y = y_0 + v_0 t + \frac{1}{2} a_c t^2$$

$$v^2 = v_0^2 + 2a_c(y - y_0)$$